

KeynetikPOS

System Documentation

Installation · Architecture · Administration · Operations

Version 2.0.0

Keynetik Solutions · Qt 6 / C++17 · Windows

About this document

This document describes KeynetikPOS as built and deployed: what it requires, how it is installed and configured, where its data lives, how licensing works, and how to operate, back up, and troubleshoot it. The audience is installers, system administrators and IT support staff.

For a code-level walkthrough of every class see **CODE_DOCUMENTATION.md**; for day-to-day cashier and manager usage see the **User Guide**.

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1. System overview

KeynetikPOS is a single-machine, offline-first point-of-sale application for small and mid-size retailers, built with the Kenyan market in mind (KSh currency, 16% VAT context, M-Pesa mobile money, WhatsApp/SMS reporting). All business data lives in a local SQLite database on the terminal; the only external dependencies are optional: an online license server (activation/heartbeat) and optional messaging gateways (Twilio WhatsApp, Africa's Talking SMS) for scheduled reports.

The application is self-contained: it bootstraps its own database schema on first run, seeds a default administrator account, and needs no separate database server, web server, or runtime install beyond the bundled Qt DLLs.

Capability summary

Area	Capabilities
Sales	Product grid + search, barcode scanning, multiple carts, PIN-gated discounts, cash/card/mobile/split payments, receipt printing / PDF.
Inventory	Catalog, stock levels, low/critical alerts, restock, CSV import/export, audit-logged adjustments.
Users	RBAC (Admin/Manager/Cashier/Viewer), PBKDF2 passwords, forced password change, user-action audit log.
Reporting	Analytics dashboards, operational reports (CSV/PDF), sales history, refunds, end-of-day Z-Reports.
Shifts	Open/close shifts with counted cash floats, over/short reconciliation.
Messaging	Scheduled Z-Report / inventory / sales reports via WhatsApp or SMS.
Licensing	30-day trial, CD-key activation (online + offline fallback), device limits, revocation, tamper detection.

2. System requirements

Hardware

Item	Minimum	Recommended
CPU	Dual-core x86-64	Quad-core
RAM	2 GB	4 GB+
Disk	300 MB free + DB growth	1 GB+ free
Display	1024 x 640 (enforced min)	1366 x 768 or larger

Software

- **Operating system:** Windows 10 or 11 (64-bit). Verified on Windows 11.
- **Runtime:** none to install separately — the installer bundles all required Qt 6 runtime DLLs via windeployqt. No system-wide Qt install is needed on the target.

Optional peripherals

- **Barcode scanner** — USB keyboard-wedge (HID, no driver) or serial / virtual-COM (RS-232). Both supported (see §7).
- **Printer** — any Windows printer; receipts can also export as PDF with no printer at all.

Network

- **Not required for normal operation** — the POS runs fully offline.
- Internet is used only for (a) license activation/heartbeat and (b) sending scheduled WhatsApp/SMS reports, if configured. Offline terminals fall back to local license validation and skip report sending.

3. Architecture

3.1 Process & startup sequence

KeynetikPOS is a single-process desktop application. **main()** runs a series of sequential gates at startup, each protecting the next:

```
Anti-tamper -> License validation -> Database init -> Login (max 3)
-> Forced password change (if required) -> MainWindow
-> (deferred ~2s) online license heartbeat
```

- **Anti-tamper** — light, debugger-only check on release builds; generic error and exit if a debugger is attached.
- **License validation** — strictly offline and non-blocking at startup (§6). Trial/expired/invalid states prompt for a CD key.
- **Database init** — opens/creates the SQLite DB, builds schema, seeds sample products and the default admin on a fresh install.
- **Login** — capped at 3 attempts; a must-change-password account is forced through a password change before the main window opens.
- **Deferred heartbeat** — online revalidation runs ~2s after the UI shows, so a slow/absent network never freezes startup.

Logout does **not** restart the process — it closes the main window and loops back to the login dialog within the same process.

3.2 Layered design

- **Singletons** for process-wide services: Database, UserManager, LicenseManager, ThemeManager — one instance, one source of truth.
- **Managers** (QObject) hold business logic + persistence; **Dialogs** (QDialog) hold UI only and delegate to managers.
- **Self-bootstrapping schema** — each manager creates its own tables on construction and adds missing columns via a lightweight ensureColumn() migration.
- **Parameterised SQL everywhere** — prepared statements with bound values (SQL-injection safe).
- Single-threaded DB access; SQLite runs in WAL mode with a busy timeout for resilience.

3.3 Technology stack

Layer	Technology
Language	C++17
UI / framework	Qt 6 (Widgets, Core, Sql, PrintSupport, Network, SerialPort)
Database	SQLite (via Qt SQL, WAL journal mode)
Build	CMake >= 3.16, MinGW or MSVC toolchain
Packaging	CPack + NSIS installer, windeployqt for runtime DLLs
License server	Cloudflare Worker + D1 (separate, in license-server/)

4. Data model & storage

4.1 Where data lives

Data	Location
Business database	pos_database.db in the per-user AppData directory (QStandardPaths::AppDataLocation), falling back to the executable directory.
License state	Windows registry HKCU\Software\KeynetikSolutions\KeynetikPOS\License, plus a machine-bound breadcrumb file in the user config dir.
Theme / receipt / messaging keys	QSettings under org "KeynetikPOS".
Generated receipts	Receipt_<saleId>_<timestamp>.pdf (current build emits PDF).
Per-user database The database lives per Windows user under AppData, not in Program Files (read-only for non-admins). If multiple Windows accounts share the terminal, each gets its own database unless you relocate it.	

4.2 Core tables (created automatically)

Table	Purpose
products	Catalog: name, category, price, cost_price, profit_margin, stock_quantity, barcode, is_active.
sales	Transaction headers: total, tax, discount, payment_method, amount_paid, change_due, sale_date.
sale_items	Line items with snapshots of name/price/cost at sale time.
users	Accounts: username, password_hash (PBKDF2), full_name, email, role, is_active, must_change_password.
stock_adjustments	Audit log of manual stock changes (reason, who, when).
refunds	Refund records linked to the original sale.
AppSettings	Key/value business settings (currency, tax, hashed discount PIN...).
Shifts	Cashier, opening/closing float, running totals, open flag.
payments	Per-tender payment records (supports split tender).
message_schedules	Scheduled report definitions.

Key choices: sale items **snapshot** product name/price/cost so historical reports and profit math stay correct after products change; money is stored as REAL (double) by current design — see §11.

5. Installation & deployment

5.1 Installing on a terminal

The standard deliverable is an NSIS installer: **KeynetikPOS-2.0.0-Setup.exe**.

- Run the installer (admin rights recommended so shortcuts are created system-wide).
- It installs the executable plus bundled Qt runtime DLLs, creates Start-menu and desktop shortcuts, and can launch on finish.
- On first launch the app validates its license (or starts a 30-day trial), builds its database, and creates the default admin account (§8).

5.2 Building from source

Prerequisites: Qt 6.11 (MinGW or MSVC), CMake >= 3.16, a C++17 compiler. Verified Release build (MinGW, PowerShell):

```
$env:PATH = "C:\Qt\Tools\mingw1310_64\bin;" + `
"C:\Qt\6.11.0\mingw_64\bin;" + $env:PATH
cmake --build "build\Desktop_Qt_6_11_0_MinGW_64_bit-Release"
```

Required Qt components: Widgets, Core, Sql, PrintSupport, Network, SerialPort (and Test for the unit tests).

5.3 Packaging the installer

When LICENSE.txt exists, CPack produces an NSIS installer with icon, shortcuts and launch-after-install; windeployqt stages the exe with all runtime DLLs.

```
cpack -G NSIS # from the build directory
```

5.4 Tests

A QtTest target (carttotals_test) covers the pure cart-totals math (tax, discount, rounding — 15 cases). Enabled by default (-DKEYNETIK_BUILD_TESTS=ON).

```
cmake --build <build-dir> --target carttotals_test
& "<build-dir>\carttotals_test.exe" -o results.txt,txt
ctest --test-dir <build-dir>
```

6. Licensing system

KeynetikPOS uses an offline-first licensing model so a shop with no internet is never locked out, while still enforcing device limits and revocation when online.

6.1 License states & flow

State	Meaning
Trial	30-day evaluation (TRIAL_DAYS=30). Nags at first run and when ≤ 7 days remain.
FullLicense	Activated with a valid CD key.
TrialExpired	Trial used up — prompts for a CD key.
InvalidKey	Stored license invalid or revoked — prompts for a CD key.

- **Activation** is online-first (activateOnServer): the server enforces device limits and revocation. If unreachable, the app falls back to local format validation (activateKey) so offline shops can still activate; local activations are never heartbeat-checked.
- **Heartbeat** (startOnlineHeartbeat) runs ~2s after the UI is up, asynchronously revalidates server-activated keys, and tracks offline days against a 7-day grace period (MAX_OFFLINE_DAYS).

6.2 Anti-tamper

- License state in HKCU is covered by a salted SHA-256 hash; hand-editing the registry trips wasTampered() and blocks startup.
- A second machine-bound signed copy of the install date (a hidden breadcrumb file) prevents trial resets by wiping the registry alone — startup takes the earliest of the two dates.

Operational caution

Changing TAMPER_SALT or the registry-hash formula after shipping will brick legitimately installed copies and requires a migration. A local tamper lockout can be cleared by deleting the HKCU ...\\KeynetikPOS\\License key (resets activation, not the trial breadcrumb).

6.3 License server (deployed)

- Cloudflare Worker + D1 (code in license-server/, free tier).
- **Live endpoint:** <https://keynetik-license.kelvinyabate.workers.dev> (SERVER_URL in licensemanager.cpp).
- Admin ops (mint/revoke/list) require a bearer ADMIN_TOKEN (Cloudflare Worker secret); admin script license-server/admin.ps1, schema license-server/schema.sql.
- Key minting: tools/generate-keys.ps1. Keys are XXXX-XXXX-XXXX-XXXX with a checksum segment.
- Server reasons returned to the client: device_limit_reached, key_revoked, key_expired.

Wrangler & token notes

Wrangler (Cloudflare CLI) needs Node 22+. The ADMIN_TOKEN must be set with no trailing newline (a shell-pipe newline previously caused a 401; the server trims it defensively).

7. Peripheral configuration

7.1 Barcode scanners

BarcodeReader supports both common scanner types behind one signal (barcodeScanned):

- **Keyboard-wedge (USB HID)** — the scanner types the code + Enter. Installed as an app-wide Qt event filter; an inter-character timer (~100 ms) distinguishes a scan burst from human typing. No driver, no config for most scanners.
- **Serial (RS-232 / virtual COM)** — default 9600-8-N-1; available COM ports are enumerated for the settings UI. Configurable prefix/suffix stripping removes STX/ETX framing bytes.

The app can also generate and print EAN-13 labels (valid check digits) for unlabelled stock, plus Code128/QR strings — no GS1 subscription required.

7.2 Receipts & printing

ReceiptPrinter builds receipts as fixed-width text for 40-column thermal printers and as styled HTML rendered via QPrinter for standard printers and PDF export. **In the current build receipts are emitted as PDF by default** (Receipt_<saleId>_<timestamp>.pdf), usable with zero printer setup. Company header/footer and printer config persist in QSettings.

8. Administration

8.1 First-run default account

Field	Value
Username	admin
Password	admin123
Role	Admin
Flag	must_change_password = true (new password required at first login)

Change this immediately

The default password is published in source and logs. The first login forces a password change; do not skip or share it.

8.2 Roles & permissions (RBAC)

Permission	Admin	Mgr	Cashier	Viewer
Make / view sales	Yes	Yes	Yes	view
Apply discounts	Yes	Yes	Yes	-
Void / delete transactions	Yes	void	-	-
View inventory	Yes	Yes	Yes	Yes
Add/edit products, adjust stock	Yes	Yes	-	-
Delete products	Yes	-	-	-
View reports / analytics	Yes	Yes	-	Yes
Export reports	Yes	Yes	-	-
Manage users	Yes	-	-	-
View users	Yes	Yes	-	-
Settings, backup, view logs	Yes	-	-	-

8.3 Password & account policy

- Passwords hashed with salted PBKDF2-SHA256 (100,000 iterations, 16-byte OS-random salt). Legacy unsalted SHA-256 hashes are transparently rehashed on next login.
- Minimum password length 8; new passwords must differ from the old.
- Admin-set passwords are temporary (must_change_password).
- Logins, password changes and significant user actions are audit-logged.

8.4 Business settings & 8.5 Scheduled messaging

- Settings (Admin only): business identity, currency (default KSh / KES), receipt footer, tax (rate/label, inclusive pricing), receipt-print toggle, hashed discount PIN, and messaging credentials. Stored in the AppSettings table so they travel with a backup; saving triggers a live refresh.
- ScheduleManager polls once a minute and fires due schedules (Daily/Weekly/Monthly/OnShiftClose/OnSalesThreshold), generating and dispatching reports through the chosen provider. Schedules and lastSent timestamps persist; "Send Now" tests credentials immediately.

9. Backup & recovery

The entire business state is in one file: **pos_database.db** (plus the WAL/SHM sidecar files SQLite may create alongside it).

9.1 Backup procedure

- Close KeynetikPOS (checkpoints WAL; no write mid-flight).
- Locate the database in the per-user AppData directory (under ...\\AppData\\...\\KeynetikPOS).
- Copy pos_database.db (and any -wal / -shm files if present) to a safe location — external drive, network share, or cloud folder.

Because all business settings live inside the database, this single copy captures products, sales, users, settings, shifts and schedules.

9.2 Restore

Close the app, replace pos_database.db in AppData with the backup, relaunch. Don't mix a database with mismatched WAL sidecars — copy them together or checkpoint first by opening/closing cleanly.

9.3 What is not in the database

- License state (registry + breadcrumb) — re-activation may be needed on a new machine, subject to the key's device limit.
- Theme choice and printer/messaging org keys stored in QSettings.

Simplest policy

Schedule an OS-level backup of the AppData folder.

10. Security considerations

- **Authentication:** PBKDF2-hashed passwords with constant-time comparison; login attempts capped at 3 per session.
 - **Authorization:** RBAC enforced in the UI (disabling controls) and in managers (permission re-checks). Discounts and refunds are PIN/permission-gated and audit-logged.
 - **SQL injection:** prevented via prepared statements throughout.
 - **Audit trails:** user actions, stock adjustments, discounts and refunds logged with who/when.
 - **Anti-tamper / licensing:** the in-app checks are an obfuscation/format gate — the license server is the real authority; client-side TAMPER_SALT and hash are deterrents, not cryptographic security.
 - **Data at rest:** the SQLite database is not encrypted. Rely on Windows account permissions and disk encryption (BitLocker) for confidentiality.
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11. Known limitations & technical debt

- **Money stored as double/REAL**, not integer cents. Totals math is unit-tested and “contained” (a characterization test pins the rounding artifact, e.g. `roundCents(1.005) -> 1.00`), but a full integer-cents migration is deferred.
- **Single-machine, single-process, single-threaded** — no multi-terminal networking or concurrent DB writers. WAL + busy-timeout cover crash resilience, not multi-user concurrency.
- **Overlapping UIs kept in the build** — three analytics screens and two inventory screens overlap in scope; consolidation is a future cleanup.
- **Inventory reorder levels** are not persisted per-product in the database.
- **Email receipts** are a stub; WhatsApp/SMS are the live channels.
- **No built-in automated backup** — backups are a manual/OS-level procedure (§9).

12. Troubleshooting

Symptom	Likely cause	Resolution
“License data has been modified...” at startup	Registry tamper-hash mismatch (salt/formula change or manual edit)	Delete HKCU ...\KeynetikPOS\License and re-activate.
“Error code: 0x00000005” and exit	Anti-tamper tripped (debugger attached) on a release build	Close debuggers/diagnostic tools and relaunch.
Trial expired sooner than expected / won't reset	Breadcrumb holds the earliest install date (by design)	Trial resets are intentionally prevented; obtain a CD key.
Activation fails offline with “device limit”	Reached the server; key at its device cap	Free a device via admin (admin.ps1) or use another key.
Scheduled reports never send	Missing/wrong messaging credentials, or terminal offline at fire time	Use “Send Now” to test; verify provider keys in Settings.
Scans go to wrong field / nothing happens	Wedge timing, or wrong serial port	Confirm scanner mode; pick the correct COM port; check prefix/suffix.
Forgot the admin password	No self-service reset for the last admin	Restore from a pre-change backup, or reset via direct DB access.
Database “failed to initialize”	AppData not writable / disk full / locked DB	Check disk space and permissions; ensure no second instance holds the file.
Receipts not printing	PDF-by-default in current build	Find the generated Receipt_*.pdf; configure a printer for hardware printing.

Admin token storage

The deployed ADMIN_TOKEN was written to %TEMP%\knk_admin_token.txt on the deployer's machine — move it to a password manager and delete the temp copy.

13. Reference

- **Code-level documentation:** CODE_DOCUMENTATION.md (repo root).
- **End-user manual:** USER_GUIDE.md / KeynetikPOS_User_Guide.pdf (this folder).
- **Engineering / feature PDFs:** KeynetikPOS_Engineering_Documentation.pdf, KeynetikPOS_Feature_Documentation.pdf (this folder).
- **License server:** license-server/README.md.
- **Support:** support@keynetik.com.

KeynetikPOS v2.0.0 — Keynetik Solutions. This document reflects the system as built; verify version-specific details (default credentials, server URLs, key salts) against the current source before relying on them in production.